

Assessment 3 - Instructions



The Purpose

Food and activity play a huge role in your health and can contribute to certain types of metabolic disorders. By taking a closer look at our eating and activity habits we can predict if we're at risk of developing certain conditions. This allows us to take preventative measures by changing our lifestyles. The purpose of this task is to diagnose if an individual has diabetes by analysing lab test results and a 3-day food and exercise journal and reflect on your own food and activity habits. This assessment also helps you to develop experimental design, data analysis, and problem-solving skills.

The Task

Alice had been feeling unwell for a few days and her doctor has asked her to do a blood test. Her results indicate that she has high blood glucose levels and triglyceride levels. The doctor suspects that she might be at risk of developing diabetes or may have already developed diabetes. To confirm this, the doctor has asked her to do another test called the glucose tolerance test and keep a food and activity journal for 3 days. Your task is to analyse the results from the glucose tolerance test and based on the results determine if she has diabetes. You also need to look at her food and exercise journal and see if there are any red flags that might increase her risk of developing metabolic diseases and make some recommendations to change her lifestyle if necessary. You will also be asked to reflect on your own activity and eating habits and compare them to Alice.

Below are some personal statistics about Alice.

- **Age:** 25
- **Weight:** 72 kg
- **Height:** 1.59 m
- **Medical History:** No known conditions and not pregnant

During the LIVE tutorial in Week 7, Nirmani will go through this assessment task in more detail.

All the Necessary Resources

- Introduction to Assessment 2 SLIDES (TBA on Moodle in Week 7)
- Introduction to Assessment 2 RECORDING (TBA on Moodle in Week 7)
- How to use Excel for result Analysis example - [Instructional Video](#) AND [Practice Data Sheet with Formula](#)
- Alice's Food and Activity Journal - [Excel File for you to analyse](#)
- Alice's Glucose Tolerance Test Results - [Excel File with data for you to analyse](#)

Expectations and Marking Criteria

Generative AI Use: Generative AI can be used for simple editing assistance. For example, for fixing grammar or polishing your written work. If you use AI for this purpose, you must attribute this in the references. You must also keep your original drafts before editing using AI as your course coordinator may request you to submit these if the originality of your submitted work is unclear. You may also be called to a meeting to explain your understanding of the submitted content. More instructions on the assessment task can be found on Moodle.

Diagnosis and recommendations report requirements are listed below. Total marks available is 50.

- Analyse the glucose tolerance data (Excel file with data) and construct graphs to help diagnose Alice's condition. [This video](#) is a step-by-step guide on how to use Excel to do calculations and draw graphs.
- Use the food and activity data (Excel file with data) to calculate Alice's average energy intake and expenditure and macronutrient profile.
- Please note that you are not required to submit either of these files. They will be used to write the diagnosis report.

Diagnosis Report

Using the information from the two files answer the following two questions. You must answer each of the dot points to receive full marks.

- **Question 1: What is your diagnosis of Alice? (10 marks)**
 - *Include the glucose tolerance graph in your answer.*
 - *Are her sugar levels normal? If abnormal, what is your diagnosis?*
 - *How did you come to this conclusion? You need to compare Alice's glucose tolerance test values to the expected values from the literature. You should also mention the purpose of the healthy individual's samples.*
 - *Use evidence from the literature and your graph to support your conclusion.*
- **Question 2: Looking at Alice's food and activity record, what are your thoughts on her eating and activity habits? Do you recommend she make changes to her diet and activity levels? If so what kind of changes and why? The specific points below must be supported with specific examples where relevant and evidence to support your claims from the literature (28 marks)**

Health Statistics and General Health (8 marks)

- *Using the information provided in the Excel file, calculate the daily energy requirement value using the [Daily Energy Requirements Calculator](#) for Alice. Assume the physical activity level to be 1.6 (mostly sedentary work).*
- *How does this value compare to her intake values?*
- *Refer to Alice's energy expenditure values.*
- *Based on intake and expenditure values, is she gaining, losing, or maintaining weight?*
- *Should she gain, lose, or maintain weight and why?*

About the Diet (10 marks)

- *Is Alice getting all the macronutrients from her diet in sufficient amounts?*
- *How are her food choices? Provide specific examples from the journal.*
- *Are there any red flags that might cause health concerns such as the potential for developing diabetes?*
- *Is she eating enough meals? Meeting calorie requirements? Other concerns related to the diet journal?*
- *Do you have any recommended changes to the diet?*

About Activities (10 marks)

- *What type of activities does Alice carry out daily? Is it incidental or athletic? What's the difference and why does it matter?*
- *Refer to specific examples and MET values from the journal to what you consider to be good/bad activity choices and why.*
- *What type of activities contribute mostly to energy out?*
- *Is she getting enough sleep? Why does sleep contribute to a lot of energy expenditure?*
- *Do you have any recommended changes to her activities?*

- **Question 3: Reflect on your own diet and activity levels over the past three days. How does it compare to Alice's? Use specific examples from your own days referring to types of food, types of activities, and sleep habits. Explain whether it is necessary to make changes to your own diet and activity levels (10 marks)**
- **References (2 marks)**
 - *Must contain at least 6 references.*
 - *In-text references and the final reference list must be done in a consistent style.*

Specific Requirements

- The report should not be more than 4 pages in length. This page length excludes the reference list, graphs, diagrams, or images.
- For question two, you can answer each section in blue in single paragraphs combining all the different points OR you can use bullet points to answer the questions under each blue section.
- Use font size 11.
- Should be properly formatted with figures and table legends where necessary.

Tips

- You must start early. This assessment will take a substantial amount of time to complete (minimum 10-20 hours).
- Use Excel to do the calculations. Otherwise, your values are not going to be as accurate. When a calculation has multiple steps, you need to use unrounded values for the subsequent steps. It is much easier to do this in Excel than manually on a calculator.
- Keep an eye on the units when doing the calculations on the Excel file. Make sure you use the correct units when doing the calculations or you will end up getting the incorrect values.
- Don't get confused between calories, Calories, and kcal. [This link](#) provides a good explanation of the similarities and differences between those terms.
- You will be asked to describe your energy intake and expenditure in both kcal and kJ. Remember that 1 kcal = 4.184 kJ.

- Most of the information to support your answers will be in the lectures after Week 5. You can use lectures as references. Do not copy information from the lectures. You should be presenting the information from the lectures in your own words to support your answers.
- You should also use other sources of information such as journal articles or websites to support your answers.
- Your work will be submitted through Turnitin where we will be checking for plagiarism. If you are not sure about what constitutes plagiarism, I highly recommend you visit [this site](#) here, familiarise yourself with the information available, and complete the plagiarism lesson if you haven't done so already. Any submitted work that shows signs of plagiarism may fail the assessment.